EPSR-test-ex15 Container forces (rmaxex / pcoefexmax)

Program(s): epsr | chained passes: 2 | verification: ALL IDENTICAL (all 2 passes)

What this case tests

Exercises the container-force module of the `epsr` engine of EPSR [1,2]: a pair line in the `.pcof` file with five numbers (`rminex pcoefex coordex rmaxex pcoefexmax`) attaches every atom of one type to the nearest pseudo (Q) site of another type once it is further than `rmaxex` away. The case checks reading and writing of these lines, the extra energy term in the Monte Carlo acceptance, and the feedback of the number of escaped atoms into the force amplitude between passes.

Code exercised

Fortran (original, fortran_src/):

- `containerforces_mod.f90`: `get_containerforce_list`, `en_containerforce`, `dist_containerforce`
- `read_and_save_files.f`: `read_pcof` (5-number pair lines), `save_files_others` (writing them back)
- `epsr_standalone_rev3.f`: `generate_rp` (builds the container lists), `repulsivept` (`pcoefexmax := potexmax`)
- `ato_manipulation_rev4.f`: `fmolec1` (container energy per molecule), `atogen` (count of atoms beyond `rmaxex`); `molfit.f`: `en_change`

Rust (port, rust/epsr-core/src/):

- `containerforces.rs` (new): `ContainerForces`, `get_containerforce_list`, `en_containerforce`, `dist_containerforce`
- `pcof.rs`: `read_pcof`, `write_pcof_pairs`; `potential.rs`: `generate_rp`, `generate_mp`, `repulsivept`
- `energy.rs`: `fmolec1`, `en_change`; `structure.rs`: `atogen`

Physical system

The EPSR-test system: a liquid mixture of 200 dimethyl sulphoxide (DMSO, (CH3)2SO) and 400 water molecules (DMSO mole fraction 1/3; 3200 atoms) at 300 K in a cubic box of side 33.12 Å (atomic number density 0.0881 atoms/Å^3). DMSO uses EPSR's Lennard-Jones + Coulomb reference potential with methyl rotational groups; water uses SPC/E Lennard-Jones and charge parameters [3] with EPSR's flexible intramolecular geometry (O-H 0.976 Å). The empirical potential is refined against three neutron diffraction data sets of deuterated DMSO in D2O, in null-scattering H2O/D2O and in H2O (ISIS SANDALS runs SLS03863, SLS03871, SLS03850), the isotope-substitution approach of Soper and Luzar [4]. For this case each DMSO molecule carries one extra, massless and uncharged Q site at its sulphur position, and the container force pulls any water oxygen further than 4 Å from the nearest DMSO Q site back towards it, with an exponential energy pcoefexmax·exp((r - rmaxex)/(expecf·sqrt(rmaxex))).

Inputs

Generated from EPSR-test by `scripts/make_ex15.py ../../../EPSR-test . 4.0 0.5`: Q site and Q atom type added to every DMSO in the `.ato` file, the seven Q-x pairs added to the `.pcof` file (28 pairs), Q-O1 given `rmaxex` = `4.0` and `pcoefexmax` = `0.5`, and `iinit 1` (the stored accumulators of EPSR-test have only 21 pairs).

How the accuracy was verified

`commands.txt` (1 program call, repeated 2 times, each pass reading the files written by the previous one) was run with the Fortran reference build (`rust/tools/build_fortran_reference.sh`: the original sources, gfortran -O2 -fno-tree-vectorize, single thread) and with the Rust `epsr-rs` on identical copies of the inputs, and every file written was compared byte for byte (`rust/tools/run_ex_test.sh`, `compare_outputs.py`; only the date line of `.out` and the wall-clock columns of `fort.35` are masked). The Fortran output is stored in `fortran_reference.tar.gz`. Result of the final regression run of the complete port: every file identical in all 2 passes.
The force is active: after pass 1 the Fortran (and the port) writes `pcoefexmax` = `98` for Q-O1, the number of water oxygens found beyond 4 Å, so pass 2 runs with a much stronger force; the container energy enters `fmolec1` and every move decision of `en_change`. The 27 files written (configuration `.ato` with the random-number state, `.pcof`, energies, g(r), S(Q), potentials, `.out`) are identical after both passes.

Notes

Fortran behaviour reproduced deliberately: `ergold_container` is only updated by `fmolec1`, never after an accepted move.

References

[1] A. K. Soper, "Empirical potential Monte Carlo simulation of fluid structure", Chem. Phys. 202, 295-306 (1996).
[2] A. K. Soper, "Partial structure factors from disordered materials diffraction data: An approach using empirical potential structure refinement", Phys. Rev. B 72, 104204 (2005).
[3] H. J. C. Berendsen, J. R. Grigera and T. P. Straatsma, "The missing term in effective pair potentials", J. Phys. Chem. 91, 6269-6271 (1987).
[4] A. K. Soper and A. Luzar, "A neutron diffraction study of dimethyl sulphoxide-water mixtures", J. Chem. Phys. 97, 1320-1331 (1992).
[5] A. K. Soper, "Empirical Potential Structure Refinement - EPSRshell: a user's guide", Technical Report RAL-TR-2011-012, STFC Rutherford Appleton Laboratory (2011).
[6] GNU Fortran (gfortran) 15.2, GCC, https://gcc.gnu.org/fortran/ - reference build with -O2 -fno-tree-vectorize.